

GENEMIND SAMPLES BATCH – 1 SAMPLES
COMPARISSION

GENEMIND BATCH 15 SAMPLES : (TOTAL SIZE = 59.1 GB)

STEP	INPUT FILE	PROCESS	TOOLS USED	DURATION OF TIME	DATA SIZE
1	FASTQ (R1, R2)	MULTIQC ANALYSIS	MULTIQC, FASTQC	2 HRS	14 MB
2	FASTQ (R1, R2)	FASTP ANALYSIS	FASTP	1 HRS	60 GB
3	TRIMMED FASTQ (R1, R2)	MULTIQC ANALYSIS	MULTIQC, FASTQC	2 HRS	14 MB
4	TRIMMED FASTQ (R1, R2)	HS METRICS ANALYSIS	SENTIEON	4 HRS	55 GB
5	TRIMMED FASTQ (R1, R2)	SECONDARY ANALYSIS	SENTIEON	5 HRS	75 GB
7	VCF, BAM AND BED	SAMPLE SEX AND COVERAGE ANALYSIS	VARSEQ VS CLINICAL COVERAGE AND SAMPLE STASTICS	3 DAYS	5 GB
8	VCF	VARAINT DEPTH ANALYSIS	VARSEQ	1 DAYS	20 GB
9	BAM AND VCF	TERTIARY ANALYSIS AND REPORT PREPARATION	VARSEQ	3 DAYS	20 GB
TOTAL				~ 8 DAYS	~ 300 GB

STEP : 1 – MULTIQC ANALYSIS BEFORE TRIMMING

- INPUT FILES : R1, R2 FASTQ FILES
- TOOL : FASTQC, MULTIQC
- TIME TAKEN: (2 HRS FOR 16 SAMPLES)
- DATA SIZE : 14 MB (FINAL OUTPUT)

FASTQC RESULT BEFORE TRIMMING:

Status Checks

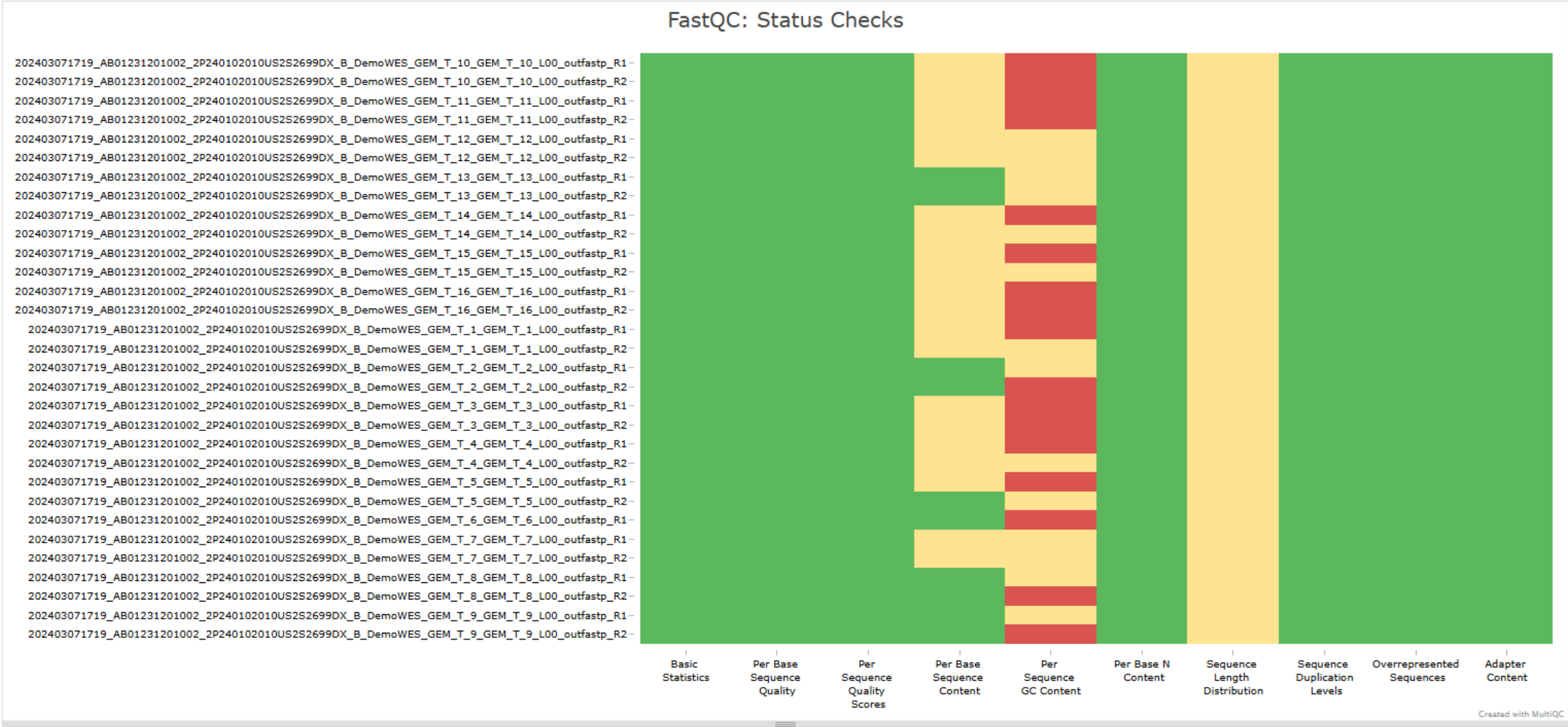
Status for each FastQC section showing whether results seem entirely normal (green), slightly abnormal (orange) or very unusual (red).

Min:

Max:

Help

Export Plot



STEP 2 : FASTP (ADAPTER TRIMMING OF FASTQ FILES)

- INPUT FILES : R1, R2 FASTQ FILES
- TOOL : FASTP
- TIME TAKEN: (1 HRS FOR 16 SAMPLES)
- DATA SIZE : 60 GB (FINAL OUTPUT)

FASTQC RESULT:

ADAPTER TRIMMING OF FASTQ FILES

BEFORE AND AFTER FASTP READS COMPARISSION:

FASTP COMPARISSION RUN 5									
BEFORE FASTP					AFTER FASTP				
Sample_ID	Sample_Name	NUMBER OF READS	SIZE OF READS	Q 30	TOTAL DATA GENERATED	NUMBER OF READS	SIZE OF READS	Q 30	TOTAL DATA GENERATED
GEM_T_1	LoveRoy	41.683354 M	150bp	92.995285%	6.252503 G	41.392676 M	143bp	93.262876%	5.954434 G
GEM_T_2	Sukanya	42.949548 M	150bp	93.021132%	6.442432 G	42.656878 M	141bp	93.284188%	6.044539 G
GEM_T_3	Vanishree	40.490188 M	150bp	92.798011%	6.073528 G	40.218506 M	144bp	93.056370%	5.796781 G
GEM_T_4	Ashifa	48.132510 M	150bp	92.702758%	7.219877 G	47.800174 M	144bp	92.937223%	6.903235 G
GEM_T_5	Bogesh	47.427120 M	150bp	92.774701%	7.114068 G	47.097536 M	145bp	93.005716%	6.829216 G
GEM_T_6	Aradhiya	42.130288 M	150bp	92.440477%	6.319543 G	41.838062 M	143bp	92.727804%	5.985110 G
GEM_T_7	Praneeth	42.477046 M	150bp	92.102598%	6.371557 G	42.177952 M	143bp	92.359557%	6.033707 G
GEM_T_8	Idhatakani	49.025776 M	150bp	93.212278%	7.353866 G	48.693306 M	139bp	93.500185%	6.814777 G
GEM_T_9	Mohamedmuzami	39.941188 M	150bp	92.259817%	5.991178 G	39.652096 M	146bp	92.468450%	5.792981 G
GEM_T_10	Supreeth	56.323712 M	150bp	93.029206%	8.448557 G	55.968490 M	144bp	93.264466%	8.099926 G
GEM_T_11	Siddanth	45.891582 M	150bp	93.633223%	6.883737 G	45.596906 M	144bp	93.867839%	6.596937 G
GEM_T_12	Kesavan	52.276000 M	150bp	93.367953%	7.841400 G	51.962706 M	139bp	93.678201%	7.261955 G
GEM_T_13	Priyankaraman	67.753738 M	150bp	92.708562%	10.163061 G	67.303144 M	144bp	92.941586%	9.756786 G
GEM_T_14	Jagadish	51.186158 M	150bp	93.537215%	7.677924 G	50.854230 M	144bp	93.812421%	7.346218 G
GEM_T_15	Muneswari	59.874758 M	150bp	93.597326%	8.981214 G	59.511012 M	141bp	93.877674%	8.433265 G
GEM_T_16	Shravansudhan	47.934772 M	150bp	93.731459%	7.190216 G	47.639772 M	140bp	94.034523%	6.671891 G

STEP : 3 – MULTIQC ANALYSIS AFTER TRIMMING

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : FASTQC, MULTIQC
- TIME TAKEN: (2 HRS FOR 16 SAMPLES)
- DATA SIZE : 14 MB (FINAL OUTPUT)

FASTQC RESULT AFTER TRIMMING:

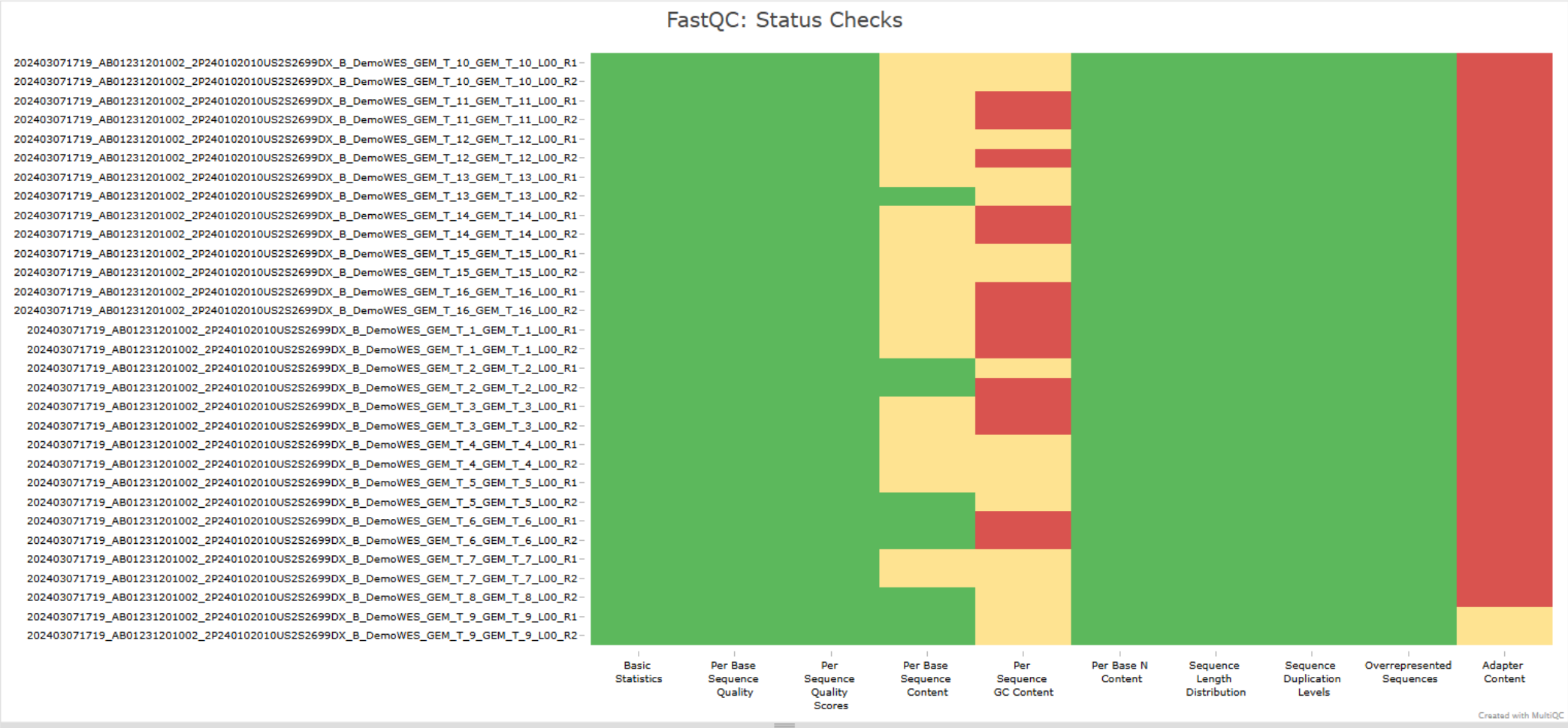
Status Checks

Help

Status for each FastQC section showing whether results seem entirely normal (green), slightly abnormal (orange) or very unusual (red).

Min: Max:

Export Plot



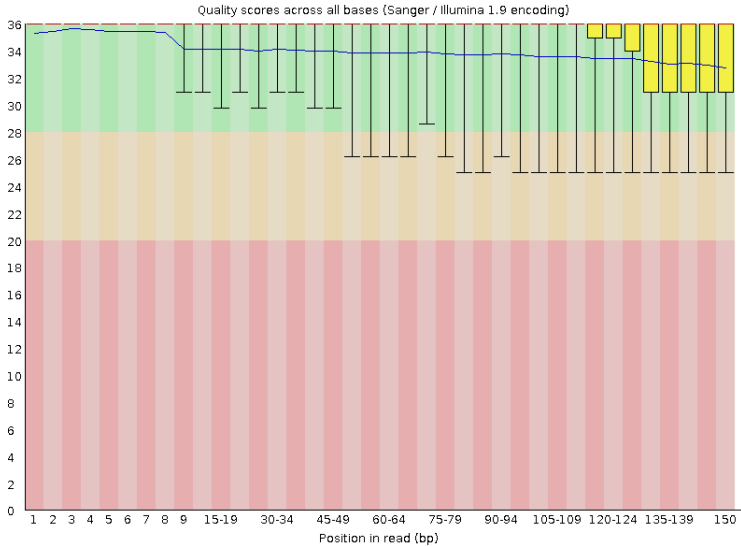
FASTQC RESULTS COMPARISSION = BEFORE AND AFTER FASTP(2 SAMPLES)

R1

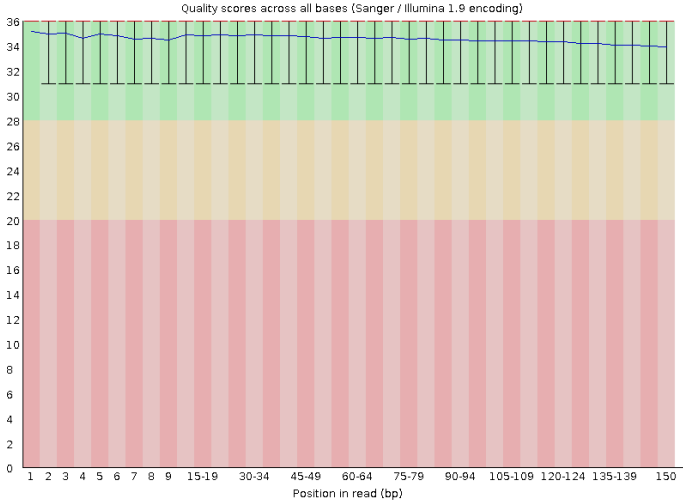
R2

BEFORE:

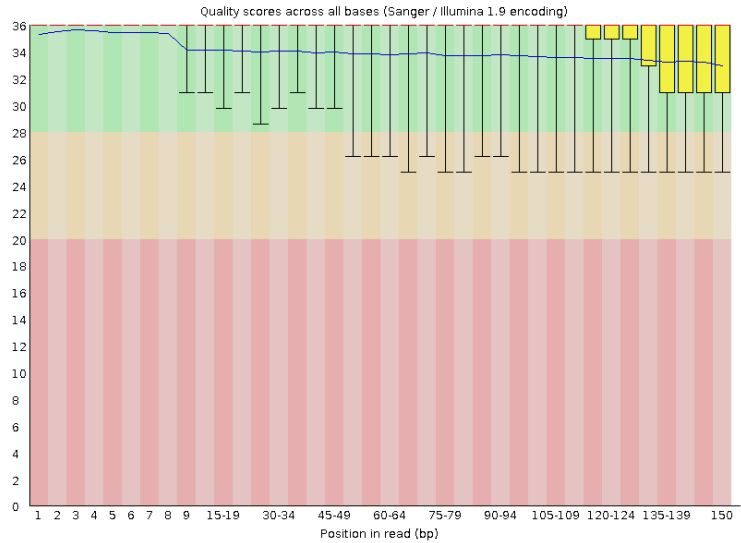
✔ Per base sequence quality



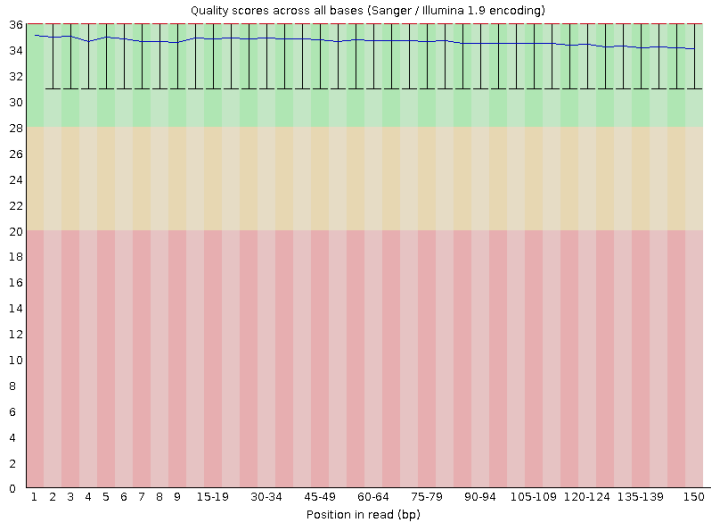
✔ Per base sequence quality



✔ Per base sequence quality



✔ Per base sequence quality



AFTER:

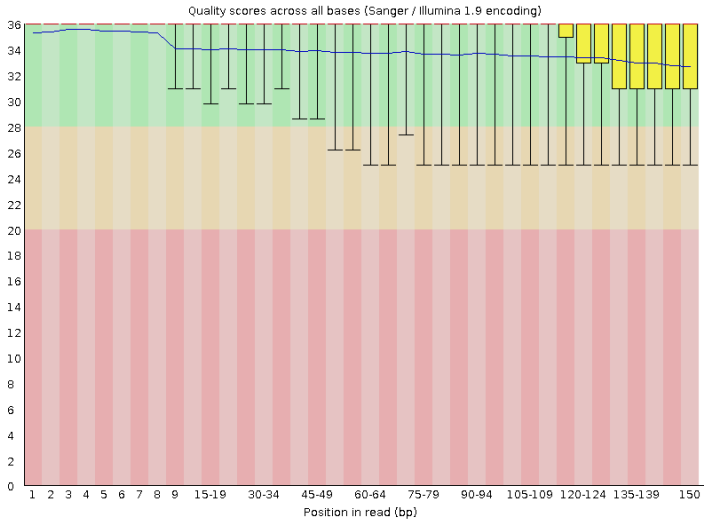
BATCH -1 = 202403071719_AB01231201002_2P240102010US2S2699DX_B_DemoWES_GEM_T_15_GEM_T_15_L00

R1

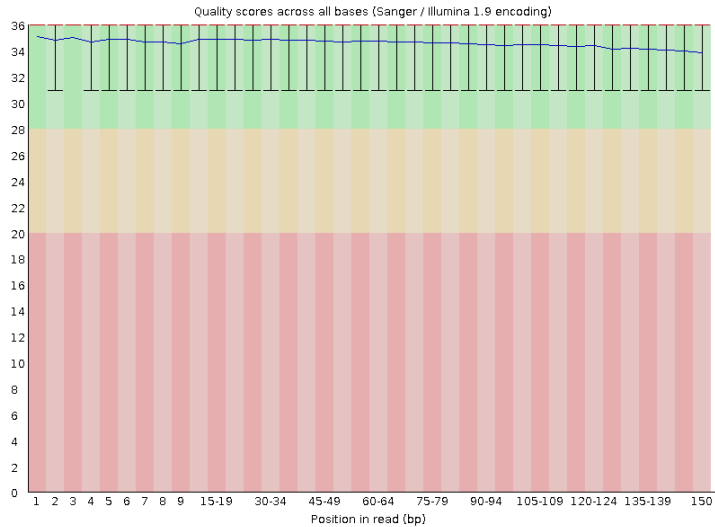
R2

BEFORE:

✔ Per base sequence quality

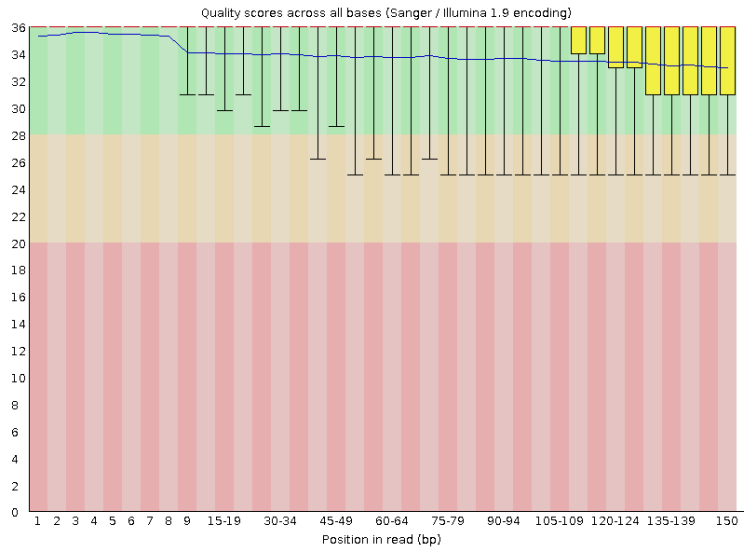


✔ Per base sequence quality

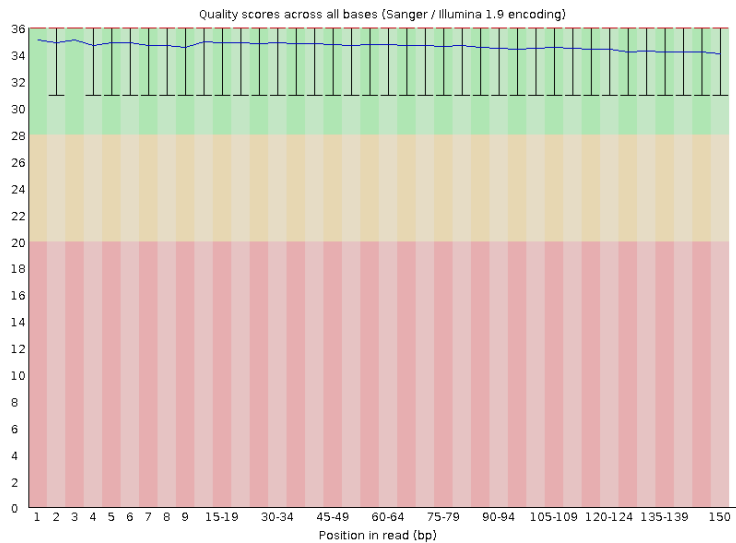


AFTER:

✔ Per base sequence quality



✔ Per base sequence quality



STEP 4 : HS METRICS ANALYSIS :

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : SENTIEON
- TIME TAKEN: (4 HRS FOR 16 SAMPLES)
- DATA SIZE : 55 GB (FINAL OUTPUT)

HS METRICS GENEMIND BATCH – 1 (GEM_T_1 – GEM_T_8):

GENEMIND BATCH – 1 TWIST BED FILE								
MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000								
BAIT_SET	BAIT							
	GEM_T_1	GEM_T_2	GEM_T_3	GEM_T_4	GEM_T_5	GEM_T_6	GEM_T_7	GEM_T_8
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815
TARGET_TERRITORY	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262
BAIT_DESIGN_EFFICIENCY	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969
TOTAL_READS	41392676	42949548	40218506	47800174	47097536	41838062	42177952	48693306
PF_READS	41392676	42949548	40218506	47800174	47097536	41838062	42177952	48693306
PF_UNIQUE_READS	41392676	42949548	40218506	47800174	47097536	41838062	42177952	48693306
PCT_PF_READS	1	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	41390477	42942893	40217370	47798466	47095796	41836783	42176686	48691516
PCT_PF_UQ_READS_ALIGNED	0.999947	0.999845	0.999972	0.999964	0.999963	0.999969	0.99997	0.999963
PF_BASES_ALIGNED	5949631029	6084050760	5795242410	6900417144	6827139898	5983712387	6030738008	6809667560
PF_UQ_BASES_ALIGNED	5949631029	6084050760	5795242410	6900417144	6827139898	5983712387	6030738008	6809667560
ON_BAIT_BASES	3379240777	3482811067	3274206096	3926208984	3806683328	3380201987	3390910948	4033035590
NEAR_BAIT_BASES	1791817610	1813817694	1779084263	2098380668	2134140496	1837175704	1865363380	1918737043
OFF_BAIT_BASES	778572642	787421999	741952051	875827492	886316074	766334696	774463680	857894927
ON_TARGET_BASES	2159178852	2177418318	2110297681	2529716533	2471591016	2224305671	2235779625	2694997168
PCT_SELECTED_BASES	0.869139	0.870576	0.871972	0.873076	0.870178	0.87193	0.871581	0.874018
PCT_OFF_BAIT	0.130861	0.129424	0.128028	0.126924	0.129822	0.12807	0.128419	0.125982
ON_BAIT_VS_SELECTED	0.653491	0.657552	0.647935	0.651697	0.640767	0.647874	0.645117	0.677619
MEAN_BAIT_COVERAGE	78.1331	80.5278	75.7045	90.7798	88.0162	78.1553	78.4029	93.2498
MEAN_TARGET_COVERAGE	59.2233	59.7236	57.8826	69.3866	67.7923	61.0096	61.3244	73.9201
PCT_USABLE_BASES_ON_BAIT	0.567517	0.540605	0.564832	0.568749	0.557411	0.564769	0.561995	0.591807
PCT_USABLE_BASES_ON_TARGET	0.362617	0.337981	0.364046	0.366454	0.361914	0.37164	0.370548	0.395464
FOLD_ENRICHMENT	42.2516	42.5844	42.0289	42.3264	41.4784	42.0229	41.8273	44.0575
ZERO_CVG_TARGETS_PCT	0.00621958	0.00684295	0.00699086	0.00694156	0.00600475	0.00679716	0.00585683	0.00708243
PCT_EXC_DUPE	0	0	0	0	0	0	0	0
PCT_EXC_MAPQ	0.0355066	0.0370936	0.0349206	0.0359175	0.0356819	0.0349634	0.035354	0.0390163
PCT_EXC_BASEQ	0.00741047	0.00739103	0.00756998	0.0079982	0.00783999	0.00813019	0.00857542	0.00758728
PCT_EXC_OVERLAP	0.285864	0.28375	0.274207	0.276647	0.261668	0.272687	0.269155	0.309937
PCT_EXC_OFF_TARGET	0.768942	0.754461	0.775608	0.773342	0.781755	0.790307	0.79307	0.787281
FOLD_80_BASE_PENALTY	1.48058	1.38892	1.44706	1.38773	1.44239	1.45261	1.4601	1.36889
PCT_TARGET_BASES_1X	0.99109	0.990564	0.990284	0.990274	0.99159	0.990308	0.99165	0.990042
PCT_TARGET_BASES_2X	0.990287	0.989804	0.989414	0.989495	0.990861	0.989504	0.990777	0.989234
PCT_TARGET_BASES_10X	0.986174	0.986325	0.985565	0.986368	0.987672	0.985785	0.98685	0.986203
PCT_TARGET_BASES_20X	0.971878	0.97846	0.974256	0.981493	0.980405	0.976276	0.975106	0.982013
PCT_TARGET_BASES_30X	0.919241	0.943218	0.925562	0.964932	0.954072	0.937291	0.931988	0.969845
PCT_TARGET_BASES_40X	0.800632	0.844508	0.803551	0.912935	0.886586	0.83505	0.831313	0.932869
PCT_TARGET_BASES_50X	0.634052	0.677529	0.625672	0.803661	0.764297	0.67548	0.676695	0.851275
PCT_TARGET_BASES_100X	0.0496911	0.0332473	0.035459	0.0999786	0.101794	0.0510726	0.0554	0.137075
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0
HS_PENALTY_10X	0	0	0	0	0	0	0	0
HS_PENALTY_20X	0	0	0	0	0	0	0	0
HS_PENALTY_30X	0	0	0	0	0	0	0	0
HS_PENALTY_40X	0	0	0	0	0	0	0	0
HS_PENALTY_50X	0	0	0	0	0	0	0	0
HS_PENALTY_100X	0	0	0	0	0	0	0	0
AT_DROPOUT	0.0228367	0.0258461	0.0232808	0.0253024	0.023807	0.0238075	0.0241587	0.0289338
GC_DROPOUT	9.39852	6.9352	9.03955	7.37687	8.3621	8.86372	8.51306	5.07034

HS METRICS GENEMIND BATCH – 1 (GEM_T_9 – GEM_T_16):

GENEMIND BATCH – 1 TWIST BED FILE								
MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000								
BAIT_SET		BAIT						
	GEM_T_9	GEM_T_10	GEM_T_11	GEM_T_12	GEM_T_13	GEM_T_14	GEM_T_15	GEM_T_16
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815
TARGET_TERRITORY	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262
BAIT_DESIGN_EFFICIENCY	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969
TOTAL_READS	39652096	55968490	45596906	51962706	67303144	50854230	59511012	47639772
PF_READS	39652096	55968490	45596906	51962706	67303144	50854230	59511012	47639772
PF_UNIQUE_READS	39652096	55968490	45596906	51962706	67303144	50854230	59511012	47639772
PCT_PF_READS	1	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	39650765	55966290	45595455	51961012	67300644	50852402	59508543	47638063
PCT_PF_UQ_READS_ALIGNED	0.999966	0.999961	0.999968	0.999967	0.999963	0.999964	0.999959	0.999964
PF_BASES_ALIGNED	5789897305	8093922910	6593932491	7251488365	9749826458	7339214521	8410262398	6660170858
PF_UQ_BASES_ALIGNED	5789897305	8093922910	6593932491	7251488365	9749826458	7339214521	8410262398	6660170858
ON_BAIT_BASES	3215030104	4836581291	3886712512	4484153239	5533661543	4330017183	5206308155	4133749724
NEAR_BAIT_BASES	1937190928	2402611597	1991878721	2005625723	3118277366	2224539008	2311736254	1833686684
OFF_BAIT_BASES	637676273	854730022	715341258	761709403	1097887549	784658330	892217989	692734450
ON_TARGET_BASES	2087969337	3013008638	2455709822	2933503262	3539252768	2732441535	3295911257	2697342031
PCT_SELECTED_BASES	0.889864	0.894399	0.891515	0.894958	0.887394	0.893087	0.893913	0.895988
PCT_OFF_BAIT	0.110136	0.105601	0.108485	0.105042	0.112606	0.106913	0.106087	0.104012
ON_BAIT_VS_SELECTED	0.624009	0.668111	0.661164	0.690956	0.639586	0.660612	0.692508	0.692718
MEAN_BAIT_COVERAGE	74.3363	111.829	89.8666	103.68	127.946	100.116	120.378	95.5784
MEAN_TARGET_COVERAGE	57.2701	82.6427	67.3567	80.462	97.0768	74.9471	90.4023	73.9844
PCT_USABLE_BASES_ON_BAIT	0.554987	0.597114	0.589169	0.617486	0.56716	0.589421	0.617354	0.619577
PCT_USABLE_BASES_ON_TARGET	0.360431	0.37198	0.37225	0.403955	0.362748	0.371952	0.390823	0.404285
FOLD_ENRICHMENT	41.3074	44.4522	43.8482	46.001	42.2211	43.8888	46.0505	46.1713
ZERO_CVG_TARGETS_PCT	0.00598714	0.00651894	0.00643089	0.0067232	0.00577583	0.00639919	0.0077375	0.00658233
PCT_EXC_DUPE	0	0	0	0	0	0	0	0
PCT_EXC_MAPQ	0.0334177	0.0332334	0.034074	0.0356258	0.0323975	0.0337195	0.0349252	0.0349098
PCT_EXC_BASEQ	0.00848193	0.00775773	0.00680597	0.00761445	0.00795313	0.00689344	0.00730639	0.00714864
PCT_EXC_OVERLAP	0.236198	0.303576	0.292897	0.332938	0.262476	0.289857	0.337232	0.335611
PCT_EXC_OFF_TARGET	0.791558	0.75145	0.760666	0.775688	0.778872	0.759867	0.753138	0.773031
FOLD_80_BASE_PENALTY	1.50711	1.50259	1.53084	1.54735	1.56576	1.49894	1.48201	1.47969
PCT_TARGET_BASES_1X	0.991627	0.990806	0.990934	0.990563	0.991748	0.990843	0.989234	0.990712
PCT_TARGET_BASES_2X	0.990733	0.990045	0.990121	0.9897	0.991071	0.990047	0.988406	0.98983
PCT_TARGET_BASES_10X	0.986402	0.987122	0.986429	0.986344	0.988235	0.986764	0.985683	0.986364
PCT_TARGET_BASES_20X	0.970102	0.981934	0.976432	0.979786	0.984359	0.980213	0.982234	0.979076
PCT_TARGET_BASES_30X	0.910328	0.965961	0.940857	0.959088	0.974446	0.958212	0.97345	0.955696
PCT_TARGET_BASES_40X	0.778091	0.926211	0.854887	0.909008	0.950104	0.902747	0.948577	0.898477
PCT_TARGET_BASES_50X	0.600537	0.849713	0.720944	0.821533	0.90145	0.802001	0.8938	0.797111
PCT_TARGET_BASES_100X	0.0387319	0.292746	0.129648	0.277678	0.436884	0.205917	0.368836	0.189909
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0
HS_PENALTY_10X	0	0	0	0	0	0	0	0
HS_PENALTY_20X	0	0	0	0	0	0	0	0
HS_PENALTY_30X	0	0	0	0	0	0	0	0
HS_PENALTY_40X	0	0	0	0	0	0	0	0
HS_PENALTY_50X	0	0	0	0	0	0	0	0
HS_PENALTY_100X	0	0	0	0	0	0	0	0
AT_DROPOUT	0.0192872	0.0200751	0.0203579	0.0220665	0.0180146	0.0205673	0.021698	0.0258462
GC_DROPOUT	10.3887	11.1178	11.105	11.403	12.4348	10.7702	10.9709	9.7427

STEP 5 : SECONDARY ANALYSIS (BAM AND VCF GENERATION):

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : SENTIEON
- TIME TAKEN: (5 HRS FOR 16 SAMPLES)
- DATA SIZE : 75 GB (FINAL OUTPUT)

SECONDARY ANALYSIS RESULT:

VCF AND BAM FILES

STEP 7 : VS CLINICAL SAMPLE AND COVERAGE STASTICS:

- INPUT FILES : VCF, BAM, BED FILES
- TOOL : VARSEQ
- TIME TAKEN: (3 DAYS 16 SAMPLES)
- DATA SIZE : 20 GB (FINAL OUTPUT)

SAMPLE STASTISTICS: SAMPLE(1 – 4)

	GEM_T_1		GEM_T_2		GEM_T_3		GEM_T_4	
	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED
Transition Count	170008	129724	171415	94441	162498	136549	175106	156335
Transversion Count	77959	59200	77131	42815	73524	64208	79383	73589
Ti/Tv	2.180736028	2.191283784	2.222387886	2.205792362	2.210135466	2.126666459	2.205837522	2.124434358
Coding Transition Count	21191	21862	22151	22050	21354	21082	21457	20619
Coding Transversion Count	7800	8230	8150	8285	7963	7803	8044	7663
Coding Ti/Tv	2.716794872	2.656379101	2.71791411	2.661436331	2.681652643	2.701781366	2.667454003	2.690721649
Singleton Count	299299	217858	299597	157834	284715	233128	307638	265795
Het Count	116210	103106	124560	83893	117503	130456	122815	134751
Het Ratio	0.388273933	0.473271581	0.415758502	0.531526794	0.412703932	0.559589582	0.399219212	0.506973419
Hom Count	183089	114752	175037	73941	167212	102672	184823	131044
Hom Ratio	0.61266564	0.527774967	0.585218173	0.469246195	0.588316838	0.441664588	0.601815642	0.49415509
Hemi Count	0	0	0	0	0	0	0	0
Hemi Ratio	0	0	0	0	0	0	0	0
SNV Het Count	96648	88682	103801	72447	97845	110230	101957	114153
SNV Hom Count	151172	100068	144580	64668	137997	90215	152343	115423
SNV Het/Hom Ratio	0.639324743	0.886217372	0.717948541	1.120291334	0.709037153	1.221858893	0.6692595	0.988996994
Indel Het Count	18644	13560	19759	10926	18670	18902	19800	19384
Indel Hom Count	31917	14684	30457	9273	29215	12457	32480	15621
Indel Het/Hom Ratio	0.584140113	0.9234541	0.648750698	1.178259463	0.63905528	1.517379786	0.609605911	1.240893669
Var Count	299299	217858	299597	157834	284715	233128	307638	265795
Multiallelic Site Count	459	432	500	260	494	662	529	607
Multiallelic Het Count	459	432	500	260	494	662	529	607
Coding Var Count	30176	31355	31521	31671	30537	29991	30698	29360
Coding Multiallelic Site Count	27	36	40	34	37	47	30	33
Coding Multiallelic Het Count	27	36	40	34	37	47	30	33
Ref Count	0	0	0	0	0	0	0	0
Ref Ratio	0	0	0	0	0	0	0	0
Called Genotypes	299299	217858	299597	157834	284715	233128	307638	265795
Call Rate	1	1	1	1	1	1	1	1
SNP Count	247918	188866	248491	137209	235962	200653	254426	229808
MNP Count	0	0	0	0	0	0	0	0
Ins Count	21560	13337	21597	9706	20383	14485	22040	16151
Del Count	29548	15424	29226	10788	28076	17620	30868	19513
DelIns Count	0	0	0	0	0	0	0	0
Complex Count	273	231	283	131	294	370	304	323
Chrom X Het Ratio	0.117204048	0.143755898	0.347845207	0.48722467	0.358958838	0.541043862	0.325540101	0.405489846
Inferred Gender	Male	Male	Female	Female	Female	Female	Female	Female
Q20 Values	5.825884 G (97.841102%)	12.380464 G (98.031999%)	5.916200 G (97.876776%)	4.902472 G (98.465368%)	5.669858 G (97.810445%)	8.866446 G (97.207834%)	6.745985 G (97.722076%)	9.123243 G (97.445939%)
Q30 Values	5.553277 G (93.262876%)	11.891220 G (94.158028%)	5.638600 G (93.284188%)	4.757541 G (95.554452%)	5.394274 G (93.056370%)	8.427033 G (92.390300%)	6.415675 G (92.937223%)	8.701137 G (92.937400%)

SAMPLE STASTISTICS: (SAMPLE 5-8)

	GEM_T_5		GEM_T_6		GEM_T_7		GEM_T_8	
	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED
Transition Count	178717	176102	165381	198396	168178	184626	172629	
Transversion Count	81738	84446	75658	96168	76430	88667	78252	
Ti/Tv	2.186461621	2.085380006	2.18590235	2.063014724	2.200418684	2.082240292	2.206065021	
Coding Transition Count	21536	21216	21455	21296	21153	21012	21112	
Coding Transversion Count	8079	7899	8020	7919	7835	7798	7884	
Coding Ti/Tv	2.665676445	2.685909609	2.675187032	2.689228438	2.699808551	2.694537061	2.677828513	
Singleton Count	315028	300773	290645	343868	294711	316895	304215	
Het Count	127156	133438	120466	161404	120821	140150	115305	
Het Ratio	0.403633963	0.443650195	0.414478143	0.469377784	0.409964338	0.442260055	0.379024703	
Hom Count	187872	167335	170179	182464	173890	176745	188910	
Hom Ratio	0.59750784	0.557486007	0.586486403	0.532191545	0.591054476	0.55906132	0.622075429	
Hemi Count	0	0	0	0	0	0	0	
Hemi Ratio	0	0	0	0	0	0	0	
SNV Het Count	105371	113587	100466	134877	100706	118081	95295	
SNV Hom Count	154892	146532	140414	159213	143734	154870	155403	
SNV Het/Hom Ratio	0.68028691	0.775168564	0.715498455	0.84714816	0.700641463	0.762452379	0.6132121	
Indel Het Count	20581	18625	19044	24499	19099	20571	18934	
Indel Hom Count	32980	20803	29765	23251	30156	21875	33507	
Indel Het/Hom Ratio	0.624044876	0.895303562	0.63981186	1.053675111	0.633339966	0.940388571	0.565075954	
Var Count	315028	300773	290645	343868	294711	316895	304215	
Multiallelic Site Count	602	613	478	1014	508	749	538	
Multiallelic Het Count	602	613	478	1014	508	749	538	
Coding Var Count	30846	30229	30685	30382	30230	29982	30194	
Coding Multiallelic Site Count	54	46	42	51	34	33	44	
Coding Multiallelic Het Count	54	46	42	51	34	33	44	
Ref Count	0	0	0	0	0	0	0	
Ref Ratio	0	0	0	0	0	0	0	
Called Genotypes	315028	300773	290645	343868	294711	316895	304215	
Call Rate	1	1	1	1	1	1	1	
SNP Count	260391	260405	240986	294406	244552	273179	250820	
MNP Count	0	0	0	0	0	0	0	
Ins Count	22748	17895	20889	21751	21107	19336	22292	
Del Count	31554	22171	28506	27149	28743	23972	30786	
DelIns Count	0	0	0	0	0	0	0	
Complex Count	335	302	264	562	309	408	317	
Chrom X Het Ratio	0.131644424	0.119222097	0.365810189	0.383431612	0.118149466	0.091305463	0.317709795	
Inferred Gender	Male	Male	Female	Female	Male	Male	Female	
Q20 Values	6.675148 G (97.743991%)	8.874134 G (97.511701%)	5.846947 G (97.691547%)	13.289725 G (98.337132%)	5.886682 G (97.563273%)	12.941381 G (97.114176%)	6.673857 G (97.932150%)	
Q30 Values	6.351561 G (93.005716%)	8.469365 G (93.063977%)	5.549861 G (92.727804%)	12.827643 G (94.917965%)	5.572705 G (92.359557%)	12.289707 G (92.223907%)	6.371829 G (93.500185%)	

SAMPLE STASTISTICS: (SAMPLE 9-12)

	GEM_T_9		GEM_T_10		GEM_T_11		GEM_T_12	
	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED
Transition Count	149395	202666	162018	92206	155397	93964	150143	94134
Transversion Count	68520	97525	74883	42050	70930	42608	68280	42524
Ti/Tv	2.180312318	2.078092797	2.16361524	2.192770511	2.190850134	2.205313556	2.198930873	2.213667576
Coding Transition Count	21533	22244	21190	21251	21220	21510	20387	20813
Coding Transversion Count	7974	8422	7865	8146	7833	8152	7410	7808
Coding Ti/Tv	2.700401304	2.641177867	2.694214876	2.608765038	2.709051449	2.63861629	2.751282051	2.665599385
Singleton Count	260330	349577	288191	154214	272587	156684	265708	157691
Het Count	119876	163010	116809	80290	113117	79484	96355	72790
Het Ratio	0.460477087	0.466306422	0.405318001	0.520640149	0.414975769	0.507288555	0.362634923	0.46159895
Hom Count	140454	186567	171382	73924	159470	77200	169353	84901
Hom Ratio	0.540740113	0.535083417	0.596011796	0.480007272	0.586202714	0.493536715	0.638860596	0.539351896
Hemi Count	0	0	0	0	0	0	0	0
Hemi Ratio	0	0	0	0	0	0	0	0
SNV Het Count	99544	136727	95591	69559	93481	68846	78121	62237
SNV Hom Count	118185	163029	141118	64577	132645	67567	140098	74283
SNV Het/Hom Ratio	0.842272708	0.83866674	0.677383466	1.077148211	0.70474575	1.018929359	0.557616811	0.837836382
Indel Het Count	19160	24467	19932	10315	18540	10114	16990	9997
Indel Hom Count	22269	23538	30264	9347	26825	9633	29255	10618
Indel Het/Hom Ratio	0.860388881	1.039468094	0.658604282	1.10356264	0.691146319	1.049932524	0.580755426	0.941514409
Var Count	260330	349577	288191	154214	272587	156684	265708	157691
Multiallelic Site Count	586	908	643	208	548	262	622	278
Multiallelic Het Count	586	908	643	208	548	262	622	278
Coding Var Count	30711	31897	30222	30760	30232	30994	28946	29925
Coding Multiallelic Site Count	47	51	33	27	41	41	46	35
Coding Multiallelic Het Count	47	51	33	27	41	41	46	35
Ref Count	0	0	0	0	0	0	0	0
Ref Ratio	0	0	0	0	0	0	0	0
Called Genotypes	260330	349577	288191	154214	272587	156684	265708	157691
Call Rate	1	1	1	1	1	1	1	1
SNP Count	217853	300046	236837	134216	226260	136519	218355	136612
MNP Count	0	0	0	0	0	0	0	0
Ins Count	18143	21911	21498	9317	19639	9313	19700	9900
Del Count	24015	27114	29474	10581	26389	10724	27290	11032
DelIns Count	0	0	0	0	0	0	0	0
Complex Count	319	506	382	100	299	128	363	147
Chrom X Het Ratio	0.143053645	0.112922561	0.122802734	0.152777778	0.121729845	0.14913511	0.128456511	0.176422094
Inferred Gender	Male	Male	Male	Male	Male	Male	Male	Male
Q20 Values	5.651960 G (97.565649%)	10.361660 G (97.783918%)	7.924805 G (97.837983%)	4.479552 G (98.574592%)	6.469788 G (98.072613%)	5.387441 G (98.482209%)	7.116408 G (97.995756%)	5.630999 G (98.487317%)
Q30 Values	5.356680 G (92.468450%)	9.931967 G (93.728865%)	7.554353 G (93.264466%)	4.355882 G (95.853172%)	6.192402 G (93.867839%)	5.232642 G (95.652483%)	6.802869 G (93.678201%)	5.466556 G (95.611176%)

SAMPLE STASTISTICS: (SAMPLE 13-16)

	GEM_T_13		GEM_T_14		GEM_T_15		GEM_T_16	
	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED
Transition Count	198650	105816	154838	88633	161099	119757	147138	101934
Transversion Count	92766	47799	71024	40413	73388	55424	67095	46613
Ti/Tv	2.141409568	2.213770163	2.180079973	2.193180412	2.195168147	2.160742639	2.192980103	2.186814837
Coding Transition Count	21674	21752	20139	20225	20595	19872	21018	21527
Coding Transversion Count	8102	8293	7583	7720	7577	7306	7741	8209
Coding Ti/Tv	2.67514194	2.622935005	2.655809047	2.619818653	2.718094233	2.7199562	2.715153081	2.622365696
Singleton Count	353012	176799	273552	148066	287117	202753	260827	171413
Het Count	140007	85492	103256	68788	105947	100997	105863	85431
Het Ratio	0.396606914	0.483554771	0.377463883	0.464576608	0.36900288	0.498128264	0.405874392	0.498392771
Hom Count	213005	91307	170296	79278	181170	101756	154964	85982
Hom Ratio	0.604694934	0.517331838	0.623915998	0.536172976	0.632446528	0.503030363	0.595328467	0.502454361
Hemi Count	0	0	0	0	0	0	0	0
Hemi Ratio	0	0	0	0	0	0	0	0
SNV Het Count	114267	73700	84080	59426	85703	85881	87159	73716
SNV Hom Count	176909	79735	141629	69479	148625	89042	126906	74696
SNV Het/Hom Ratio	0.645908348	0.924311783	0.593663727	0.855308798	0.576639193	0.964499899	0.686799679	0.986880154
Indel Het Count	24220	11186	17966	8948	18928	14182	17650	11137
Indel Hom Count	36096	11572	28667	9799	32545	12714	28058	11286
Indel Het/Hom Ratio	0.670988475	0.966643623	0.626713643	0.913154404	0.581594715	1.115463269	0.629054102	0.986797803
Var Count	353012	176799	273552	148066	287117	202753	260827	171413
Multiallelic Site Count	760	303	605	207	658	467	527	289
Multiallelic Het Count	760	303	605	207	658	467	527	289
Coding Var Count	31015	31339	28813	29094	29326	28190	29904	30968
Coding Multiallelic Site Count	41	32	42	40	40	36	33	28
Coding Multiallelic Het Count	41	32	42	40	40	36	33	28
Ref Count	0	0	0	0	0	0	0	0
Ref Ratio	0	0	0	0	0	0	0	0
Called Genotypes	353012	176799	273552	148066	287117	202753	260827	171413
Call Rate	1	1	1	1	1	1	1	1
SNP Count	291336	153555	225811	128999	234434	175095	214177	148502
MNP Count	0	0	0	0	0	0	0	0
Ins Count	25922	10764	20130	8847	21772	12455	19497	10594
Del Count	35323	12330	27267	10122	30504	14963	26853	12170
DelIns Count	0	0	0	0	0	0	0	0
Complex Count	431	150	344	98	407	240	300	147
Chrom X Het Ratio	0.125894411	0.164108187	0.122802482	0.152603232	0.312662996	0.434854772	0.125034578	0.153785489
Inferred Gender	Male	Male	Male	Male	Female	Female	Male	Male
Q20 Values	9.535369 G (97.730634%)	6.768665 G (98.526834%)	7.202760 G (98.047178%)	3.914435 G (98.606061%)	8.266926 G (98.027585%)	9.057315 G (98.083435%)	6.545726 G (98.109014%)	5.531920 G (98.455847%)
Q30 Values	9.068112 G (92.941586%)	6.576353 G (95.727479%)	6.891665 G (93.812421%)	3.807481 G (95.911842%)	7.916953 G (93.877674%)	8.923023 G (96.629158%)	6.273880 G (94.034523%)	5.369168 G (95.559227%)

COVERAGE STATISTICS : SAMPLE (1-4)

	GEM_T_1		GEM_T_2		GEM_T_3		GEM_T_4	
	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED
Sample Mean Depth	70.6256	159.787	70.431	53.2456	67.6948	101.122	83.5809	105.425
Sample Mean Forward Depth	35.3105	79.803	35.2145	26.5937	33.8435	50.5461	41.7877	52.7004
Sample Mean Reverse Depth	35.3149	79.9835	35.2163	26.6517	33.851	50.5759	41.793	52.7242
Sample Mean Filtered Depth	1.67922	8.05996	1.77283	2.81038	1.5525	3.54216	1.9841	3.04444
Sample % 1x	99.132	98.2539	99.0737	97.9892	99.0529	97.5162	99.0528	96.3786
Sample % 5x	98.8967	97.9325	98.8445	97.5565	98.816	96.4127	98.8368	94.533
Sample % 20x	97.9447	97.4988	98.1753	95.4335	98	93.6459	98.683	90.9992
Sample % 50x	78.5548	96.3415	82.3898	55.5184	77.773	86.6103	90.7453	83.5382
Sample % 100x	13.619	84.4359	9.83782	2.26929	9.09599	55.6424	26.481	61.7056
Sample % 500x	0.00990442	0.204247	0.0101348	0.00854059	0.00771019	0.010162	0.0109193	0.00941565
sample Errors	?	?	?	?	?	?	?	?
Q20 Values	5.825884 G (97.841102%)	12.380464 G (98.031999%)	5.916200 G (97.876776%)	4.902472 G (98.465368%)	5.669858 G (97.810445%)	8.866446 G (97.207834%)	6.745985 G (97.722076%)	9.123243 G (97.445939%)
Q30 Values	5.553277 G (93.262876%)	11.891220 G (94.158028%)	5.638600 G (93.284188%)	4.757541 G (95.554452%)	5.394274 G (93.056370%)	8.427033 G (92.390300%)	6.415675 G (92.937223%)	8.701137 G (92.937400%)

COVERAGE STATISTICS : SAMPLE (5-8)

COVERAGE STASTICS								
	GEM_T_5		GEM_T_6		GEM_T_7		GEM_T_8	
	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED
Sample Mean Depth	81.0688	108.84	68.8683	166.592	69.9711	150.048	84.826	
Sample Mean Forward Depth	40.528	54.4149	34.4284	83.276	34.9801	75.0128	42.4071	
Sample Mean Reverse Depth	40.5406	54.425	34.4396	83.3154	34.9908	75.0354	42.4186	
Sample Mean Filtered Depth	1.86458	3.38938	1.62866	4.97229	1.66418	4.50432	2.1964	
Sample % 1x	99.1834	97.8192	99.0584	96.8209	99.1953	97.6673	99.0277	
Sample % 5x	98.975	96.715	98.8245	95.3983	98.9489	96.5307	98.8134	
Sample % 20x	98.3851	94.2001	98.0584	93.8927	98.0632	94.4677	98.3559	
Sample % 50x	87.5812	87.2489	79.6069	91.242	79.6624	89.8955	91.8753	
Sample % 100x	24.7453	62.045	9.55871	84.0543	11.3464	77.2551	27.2118	
Sample % 500x	0.0113417	0.0115619	0.00570787	0.0439869	0.00744136	0.0309434	0.0139501	
sample Errors	?	?	?	?		?		
Q20 Values	6.675148 G (97.743991%)	8.874134 G (97.511701%)	5.846947 G (97.691547%)	13.289725 G (98.337132%)	5.886682 G (97.563273%)	12.941381 G (97.114176%)	6.673857 G (97.932150%)	
Q30 Values	6.351561 G (93.005716%)	8.469365 G (93.063977%)	5.549861 G (92.727804%)	12.827643 G (94.917965%)	5.572705 G (92.359557%)	12.289707 G (92.223907%)	6.371829 G (93.500185%)	

COVERAGE STATISTICS : SAMPLE (9-12)

	GEM_T_9		GEM_T_10		GEM_T_11		GEM_T_12	
	GENEMIND BED(TWIST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED
Sample Mean Depth	68.5122	112.049	103.02	48.6746	83.1376	57.0914	95.9275	60.0025
Sample Mean Forward Depth	34.2536	56.0094	51.5076	24.3093	41.5655	28.5165	47.9625	29.9713
Sample Mean Reverse Depth	34.2583	56.039	51.5125	24.3651	41.5718	28.5746	47.9647	30.0309
Sample Mean Filtered Depth	1.43485	3.52188	2.24522	2.41985	1.85394	3.18715	2.21702	3.4302
Sample % 1x	99.189	98.1844	99.105	98.0901	99.1167	98.0417	99.0793	98.1037
Sample % 5x	98.942	97.4208	98.9003	97.6468	98.8876	97.6182	98.8545	97.6663
Sample % 20x	97.8825	96.1613	98.4468	94.729	98.1901	95.9517	98.299	96.2703
Sample % 50x	76.192	92.7817	93.0504	45.6887	85.7331	62.371	90.259	67.2437
Sample % 100x	11.9182	66.4677	50.2152	1.37513	29.618	3.60461	43.4938	4.79682
Sample % 500x	0.00355747	0.00958989	0.0144575	0.00666313	0.0107931	0.0102196	0.010442	0.0085657
sample Errors	?	?	?	?	?	?	?	?
Q20 Values	5.651960 G (97.565649%)	10.361660 G (97.783918%)	7.924805 G (97.837983%)	4.479552 G (98.574592%)	6.469788 G (98.072613%)	5.387441 G (98.482209%)	7.116408 G (97.995756%)	5.630999 G (98.487317%)
Q30 Values	5.356680 G (92.468450%)	9.931967 G (93.728865%)	7.554353 G (93.264466%)	4.355882 G (95.853172%)	6.192402 G (93.867839%)	5.232642 G (95.652483%)	6.802869 G (93.678201%)	5.466556 G (95.611176%)

COVERAGE STATISTICS : SAMPLE (13-16)

	GEM_T_13		GEM_T_14		GEM_T_15		GEM_T_16	
	GENEMIND BED(TWST v2.0.2)	KAPPA BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED	GENEMIND BED(TWST v2.0.2)	MEDGENOME BED	GENEMIND BED(TWIST v2.0.2)	KAPPA BED
Sample Mean Depth	116.81	76.6264	92.6188	41.823	110.578	140.854	88.3303	60.3572
Sample Mean Forward Depth	58.4032	38.2719	46.3081	20.884	55.2868	70.4238	44.159	30.1449
Sample Mean Reverse Depth	58.4068	38.3543	46.3105	20.9388	55.2905	70.4303	44.171	30.212
Sample Mean Filtered Depth	2.55912	4.14181	2.033	2.13501	2.56881	4.82152	2.01153	3.04733
Sample % 1x	99.201	98.1016	99.1077	98.0875	98.9441	96.2547	99.0926	98.155
Sample % 5x	99.0106	97.7333	98.8932	97.6075	98.7371	95.0243	98.8608	97.7271
Sample % 20x	98.5872	96.8988	98.3538	92.5874	98.3663	92.9437	98.2655	96.3987
Sample % 50x	94.8375	83.5589	90.4856	29.5882	94.929	89.2746	89.0432	67.8971
Sample % 100x	60.2147	19.6978	39.8111	0.537923	56.9045	76.7213	34.6983	4.95321
Sample % 500x	0.0217809	0.0149328	0.00934757	0.00437505	0.013259	0.0242487	0.00771282	0.00826121
sample Errors	?	?	?	?	?	?	?	?
Q20 Values	9.535369 G (97.730634%)	6.768665 G (98.526834%)	7.202760 G (98.047178%)	3.914435 G (98.606061%)	8.266926 G (98.027585%)	9.057315 G (98.083435%)	6.545726 G (98.109014%)	5.531920 G (98.455847%)
Q30 Values	9.068112 G (92.941586%)	6.576353 G (95.727479%)	6.891665 G (93.812421%)	3.807481 G (95.911842%)	7.916953 G (93.877674%)	8.923023 G (96.629158%)	6.273880 G (94.034523%)	5.369168 G (95.559227%)

VARIANT DEPTH COMPARISSION:

S.NO	SAMPLE	SAMPLE NAME	DEPTH(REPORTED)	VARAINT ALLELE FREQUENCY	ALLELIC DEPTH	DEPTH (GENEMIND BED(TWIST v2.0.2))	VARAINT ALLELE FREQUENCY	ALLELIC DEPTH	GENE
1	GEM_T_1	LOVE ROY – KAPPA BED	184x	1	0, 184	63X	1	0, 63	MLC1 (-)
2	GEM_T_2	Mrs. Sukanya(NON-CARRIER) - KAPPA BED							
3	GEM_T_3	VANISHRE – MEDGENOME BED	146x	0.582192	61, 85	78X	0.551282	35, 43	BMP15
			103x	0.514563	50, 53	89X	0.47191	47, 42	ATP7B
			29x	0.344828	19, 10	105X	0.485714	54, 51	DNAJC3
			144x	0.472222	76, 68	56X	0.517857	27, 29	IGHMBP2
			107x	0.551402	48, 59	55x	0.509091	27, 28	PCK2
			115x	0.530435	54, 61	56x	0.517857	27, 29	MYO18B
			98x	0.459184	53, 45	84x	0.452381	46, 38	PRKDC
4	GEM_T_4	ASHIFA – MEDGENOME BED	140x	0.514286	68, 72	71x	0.577465	30, 41	KBTBD13 (+)
			99x	0.434343	56, 43	98x	0.5	49, 49	COL6A2 (+)
			104x	0.490385	53, 51	36x	0.333333	24, 12	RYR1 (+)
5	GEM_T_5	BOGESH - MEDGENOME BED	151x	0.456954	82, 69	69x	0.594203	28, 41	DHODH
			123x	0.292683	87, 36	68x	0.441176	38, 30	EYS
			80x	0.4	80, 99	111x	0.477477	58, 53	LIFR
			102x	0.5	51, 51	44x	0.386364	27, 17	MYO18B
			98x	0.459184	53, 45	58x	0.568965	25, 33	PRKDC
6	GEM_T_6	ARADHYA - MEDGENOME BED	38x	0.578947	16, 22	7x	0.571429	3, 4	IKBKKG (+)
			265x	0.50566	131, 134	41x	0.585366	17, 24	FKRP (+)
7	GEM_T_7	PRANEETH - MEDGENOME BED	154x	0.474026	81, 73	64x	0.53125	30, 34	UGT1A1 (+)
			101x	0.574257	43, 58	73x	0.410959	43, 30	KIDINS220 (-)
8	GEM_T_8	IDHAYAKANI (DMD) - MEDGENOME BED							DMD
9	GEM_T_9	Master. Mohammed Muzammi - MEDGENOME BED	126x	1	0, 126	55x	1	0, 55	PKLR (-)
10	GEM_T_10	Mr. Supreeth - KAPPA BED	55x	0.436364	31, 24	87x	0.37931	54, 33	PAH
11	GEM_T_11	Master. Siddanth - KAPPA BED	49x	0.489796	25, 24	135x	0.444444	75, 60	CHD8
			42x	0.47619	22, 20	36x	0.472222	19, 17	AUTS2
12	GEM_T_12	Master. Kesavan T - KAPPA BED	51x	1	0, 51	60x	1	0, 60	NAGLU
13	GEM_T_13	Master. Priyavarman - KAPPA BED	96x	0.541667	44, 52	85x	0.541176	39, 46	MFN2
14	GEM_T_14	Master. Jagadish S - KAPPA BED	29x	1	0, 29	115x	1	0, 115	SACS
15	GEM_T_15	Mrs. Munesswari - MEDGENOME BED	39x	0.589744	16, 23	76x	0.526316	36, 40	DMGDH
16	GEM_T_16	Baby. Shravan Sudhan - KAPPA BED	35x	0.514286	17, 18	118x	0.491525	60, 58	ITPR1